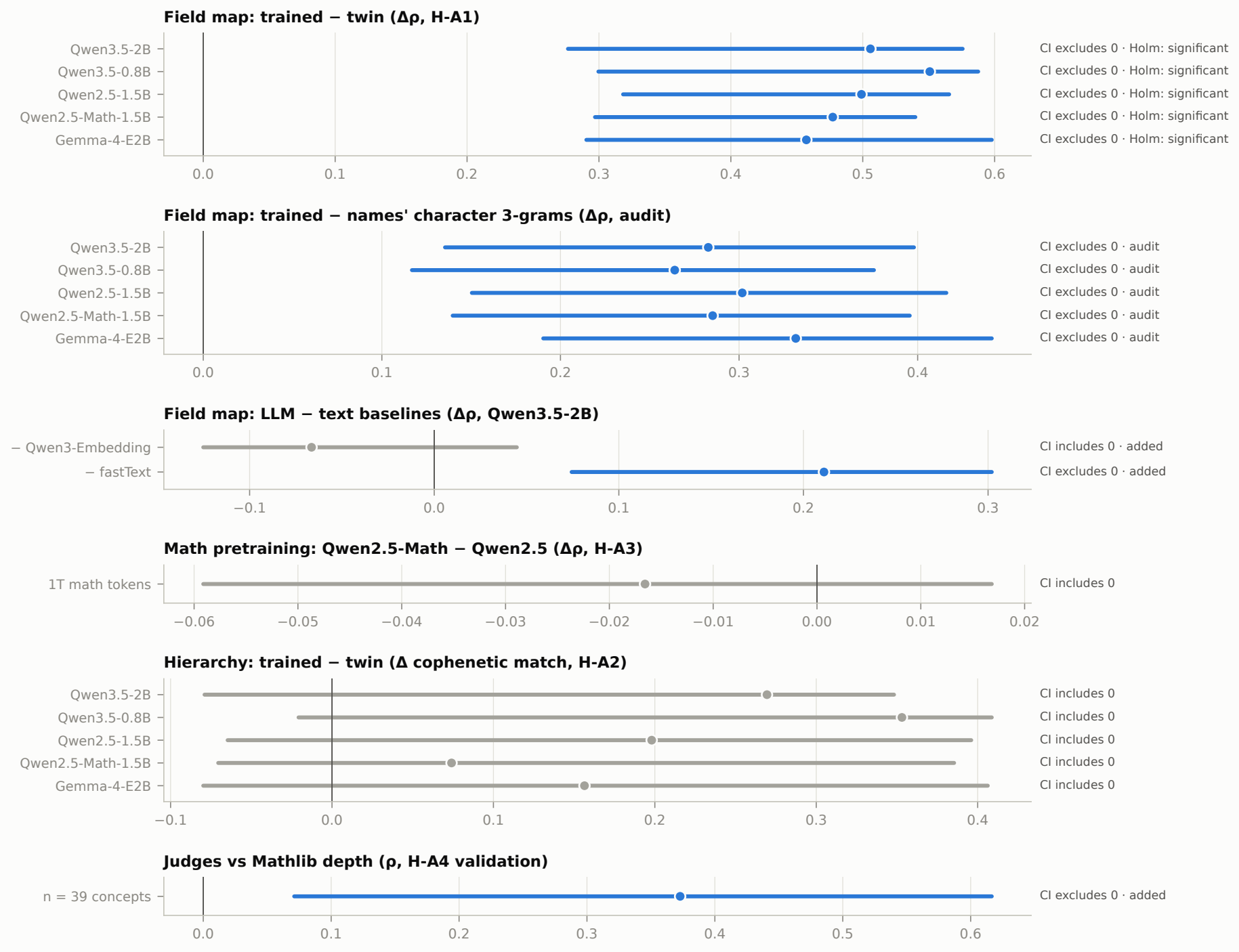
Field map: trained − twin (Δρ, H-A1)
Qwen3.5-2B
Qwen3.5-0.8B
Qwen2.5-1.5B
Qwen2.5-Math-1.5B
Gemma-4-E2B
CI excludes 0 · Holm: significant
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Field map: trained − names' character 3-grams (Δρ, audit)
CI excludes 0 · audit
Field map: LLM − text baselines (Δρ, Qwen3.5-2B)
− Qwen3-Embedding
− fastText
CI includes 0 · added
CI excludes 0 · added
−0.1
Math pretraining: Qwen2.5-Math − Qwen2.5 (Δρ, H-A3)
1T math tokens
CI includes 0
−0.06
−0.05
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
Hierarchy: trained − twin (Δ cophenetic match, H-A2)
Judges vs Mathlib depth (ρ, H-A4 validation)
n = 39 concepts